



Touchscreen in old monitors

Zmartframe turns regular monitors into touchscreens for ones who don't want to dump the old ones <http://dgit.in/Zmartframe>



BioSkin keeps buildings cool

The Urban heat island effect that keeps buildings warmer than the rural ones can be prevented with a BioSkin coating <http://dgit.in/BioSkinCoat>

Agent 001

To extend Wi-Fi range, buy a repeater or a new router?

Hi Agent 001! My house has a Wi-Fi router – the Belkin n150 F6D4230-4 version 1.802.11n router with 150mbps throughput. I've noticed that there are some black spots in the house, so a lot of rooms do not get coverage. I read about repeaters but it said that repeaters have 50% or more throughput loss because they receive a signal and throw it back on the same frequency. So should I invest in a better router or should I get a repeater? Also please let me know if it's possible for a dual band repeater to receive signal from the 2.4GHz band and throw it on the 5GHz band? And what are your product recommendations for both cases?

–A. Sanghvi

A The router that you're using, that is the Belkin N150 though an excellent value for money device is still a basic router and hence the range that it provides is less as compared to some higher end ones. And to answer your question about repeaters, well yes, repeaters do start off with a



The Netgear WN2500RP Universal Dual Band Wi-Fi Range Extender

throughput of 50% but only when they use the same radio. There are models available in the market that have two radios and use one for receiving and the other for transmitting. Then there are models which are dual band and one band

is given the job of receiving and the other does the transmitting. A popular model which does this is the Netgear WN2500RP, which can be purchased for about ₹5,000.

If you were to use this with the N150 - the single band router that you currently have, the repeated signal will be on the 5GHz band. So all the devices in the affected area need to be set to the 5GHz band. If most devices in the house aren't dual band then they'll switch to the 2.4GHz band which means you'll still experience the existing problem. So in this case, it will be prudent to go for a better router than an extender. Now comes the question of getting a better router. Now a more powerful router only does half the job, it'll be able to transmit to the farthest corners of your house but unless your devices have equally powerful antennae that can transmit all the way back to the router, it becomes a pointless exercise. If possible it would be best to buy another router and put it in another room and set it as an access point. Then use a CAT6 cable to connect it to the first router and then ensure both are transmitting on different channels. This method will work out the best but it means having an ugly CAT6 cable out in the open which other members in the household might object to. Try moving the router around the house to find a better position.

Repair old laptop or buy a new one?

Hey Agent 001, I own a Dell Studio 1558 with an Intel Core i5 chip and 4 GB of RAM. The laptop is in utter shambles: its battery has long been useless and the keyboard is broken. I'm tempted to purchase the Apple MacBook Pro 13, which is available for Rs. 64,990, and the guy at the store says that I should expect about 10k in exchange for my current laptop.

I really need a new laptop because my Dell has spent more time being fixed lately than actually doing any real work



The Apple MacBook Pro 13

and be useful. What should I do? Do I go for the Mac, do I fix my Dell or do I look at anything else with similar specs as the Dell? If you pick option 3, any suggestions?

–Niranjan

A First and foremost, Dell's Studio laptop range is an old one, and last I checked they have discontinued that line for at least 2-3 years. And if something's giving you so much pain, as your laptop apparently is by constantly being in the service centre or repair shop than actually serve any real purpose that it intended to, I think it's a good plan to just sell it off and cut your losses. Because the laptop's clearly beyond saving by mere repairs. If you're enticed to swap it for the Apple MacBook Pro, and if you want to experience life with Cupertino-based products, the offer you're getting is pretty good. I'd have said go for it, if I only knew which MacBook Pro you're getting – just ensure it's the latest model and not a year or two older, which happens very easily since Apple Macs don't have a nomenclature that reveals their version instantly.

At the same price, let's assume it to be between 50 to 55k (ballpark price of the MacBook Pro), you'd get laptops from Acer, Lenovo, Toshiba, HP, and Dell with better configuration. But remember, Apple's MacBook Pro will trump all of them in two key departments – screen quality and battery life. Also, you need to assess which operating system do you prefer from among Windows 8.1 and Mac OSX Mavericks, as that is one of the other primary factors in determining which laptop to go for. **E**